Metadata for Toppesfield (TOP), Essex, UK

Description of data set	Magnetic time-series from the BGS SWIGS differential magnetometer method (DMM) systems. Funded by NERC, grant number: NE/P017231/1 "Space Weather Impact on Ground-based Systems (SWIGS)". These data consist of measurements of the Earth's natural magnetic field at the remote site (TOPR) and the natural magnetic field plus the field created by GIC at the underline site (TOPU). The database will include .xyz files with the DMM data and one document with metadata. See Hübert et al. (2020) for further details.
Location (in lat/long): underline station	52.002528 N, 0.506644 E
Location (in lat/long): remote station	52.001681 N, 0.507867 E
Distance between DMM sites	126 m
Description of site (i.e. open moorland)	Open arable farmland close to a boundary hedge/ditch between two fields.
Equipment used	Sensys 3DFM fluxgate magnetometer and Earth Data digitisers (EDR 209, 24bit)
Installation date	28/02/2019
Removal date	24/02/2020
Number of days collected	TOPR: 361, TOPU: 363
Data sampling rate	1 Hz
Notes of known data gaps	TOPU: Data gaps: 04/09/2019, 14/09/2019. TOPR: Data gaps: 03/09/2019.
Qualitative estimate of site quality	TOPR: good, some high frequency noise in the Z component TOPU: good, strong signals in E component (not GIC related, presumably from a nearby substation)
Orientation angle of HV wires relative to true north	+4.9° (i.e. lines run approximately N-S)
Alignment of magnetometers	Magnetic north as of 28/02/2019
Voltage level of the lines	400 kV (Thornton – Lackenby)
Height above the ground of each wire (m)	Lower: 14.24 m, Middle: 23.60 m, Upper: 33.91 m
Underline magnetometer	Central (9m to either circuit)
File Type	ASCII text (.xyz)
File format	YYYY-MM-DD HH:MM:SS X Y Z Five columns with 86400 rows (plus one header row). Magnetic values in units of nT (10^-9 T). Empty values denoted by: 999999.00
Citation	Hübert, J., Beggan, C. D., Richardson, G. S., Martyn, T., & Thomson, A. W. P. (2020). Differential magnetometer measurements of geomagnetically induced currents in a complex high voltage network. Space Weather, 18, e2019SW002421. doi: 10.1029/2019SW002421